



Inspiring Excellence

Network Layer: IP Addressing

Lecture 7 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

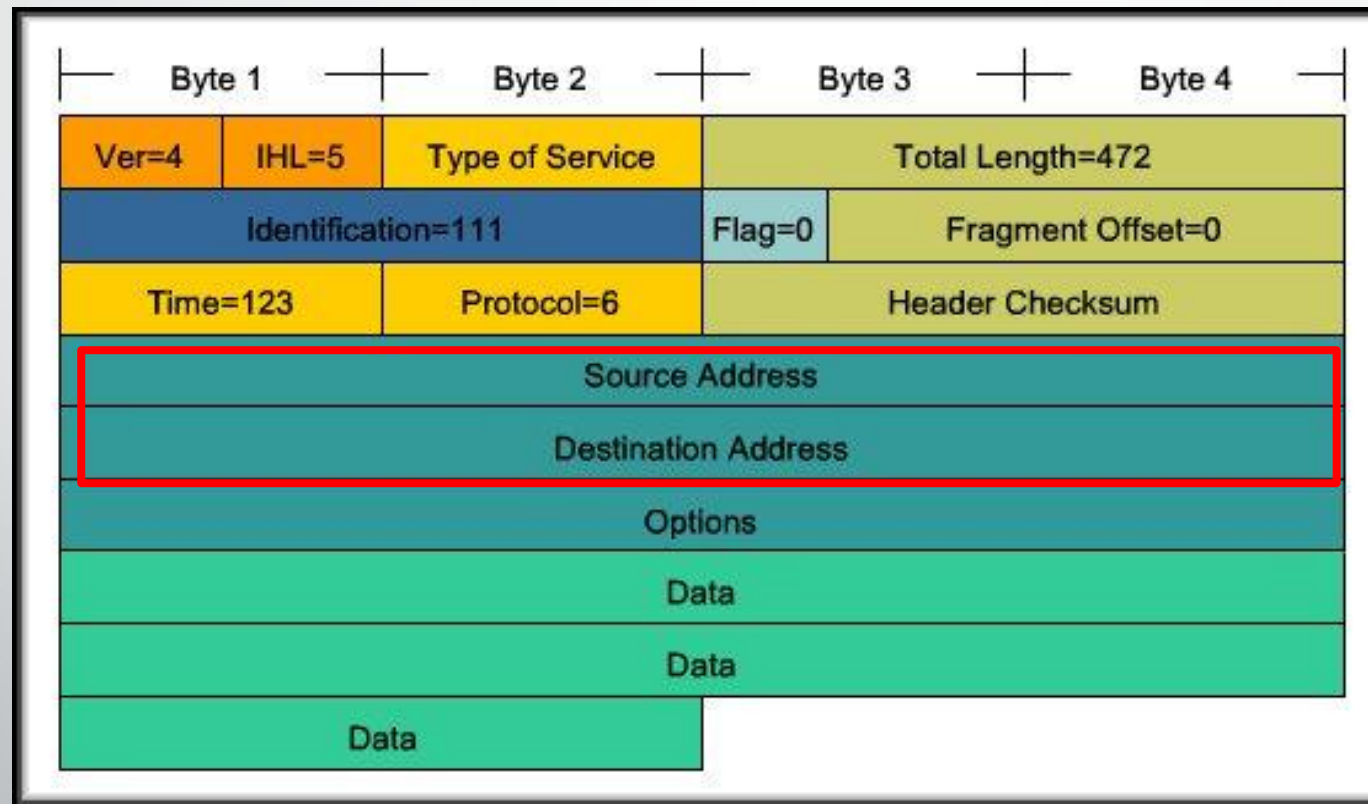
Objectives

- IPv4 Address
 - Structure
 - Subnet/Prefix Mask
- Types of IPv4 Address
- IPv6 Address
 - Structure

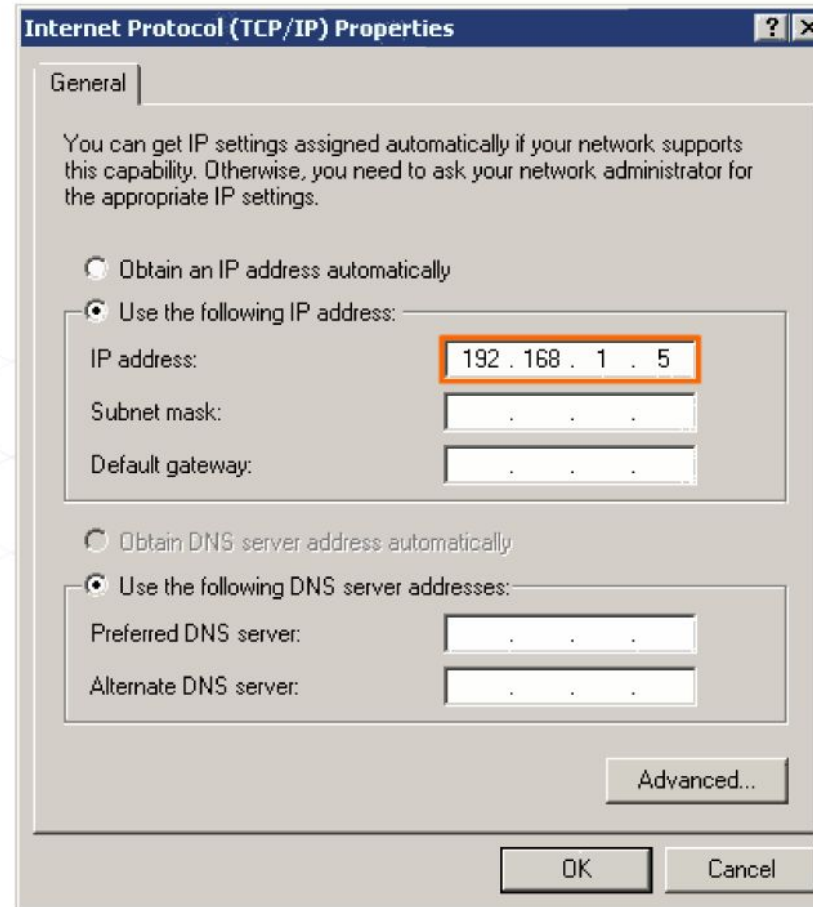
Anatomy of IPv4

Anatomy of an IPv4 Address

- Each device on a network must be uniquely identified at the Network layer.
- For IPv4, a 32 bit source and destination address is contained in each packet.



IPv4 Addressing Structure



Internet Protocol (TCP/IP) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: **192 . 168 . 1 . 5**

Subnet mask: . . .

Default gateway: . . .

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

Advanced...

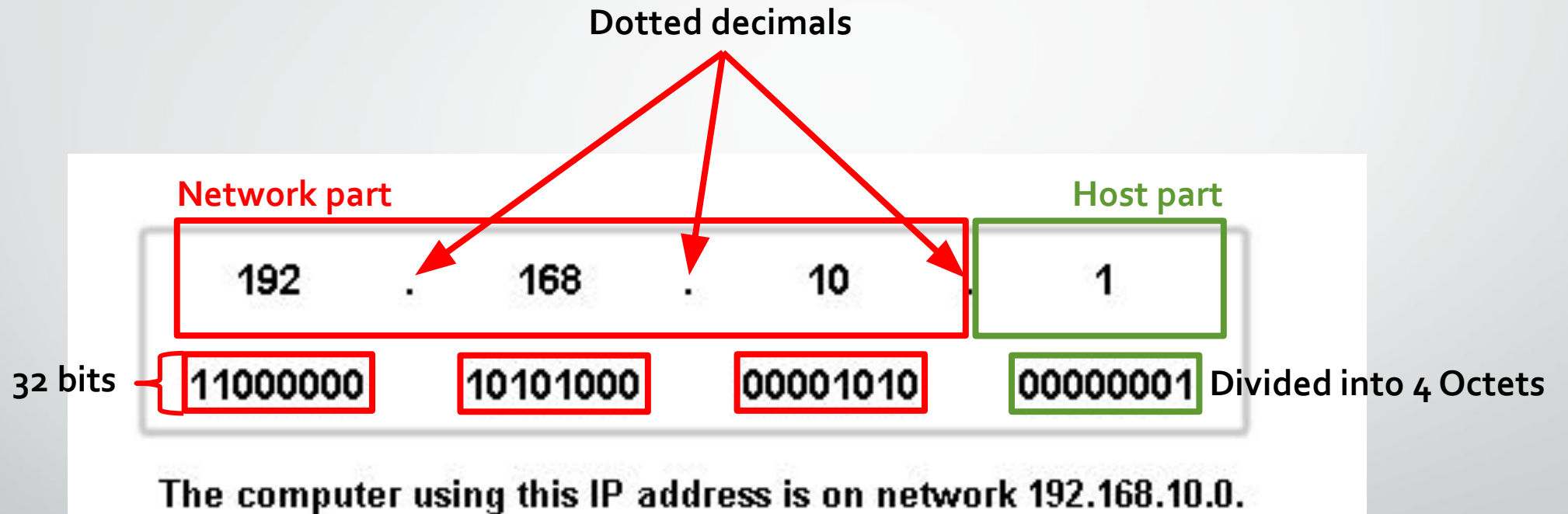
OK Cancel

I see you have
assigned me
an IP address
**11000000.1010
1000.00000001.
00000101**
Now other
hosts can find
me!



IP version 4 (IPv4) is the current form of addressing used on the Internet.

Anatomy of an IPv4 Address



Binary to decimal and Vice Versa

Binary To Decimal Conversion

Exponent	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Position	128	64	32	16	8	4	2	1
Bits	1	1	1	1	0	1	0	1
1 BYTE / 1 Octet								
Add these numbers together	128 + 64 + 32 + 16 + 0 + 4 + 0 + 1							
Decimal	245							

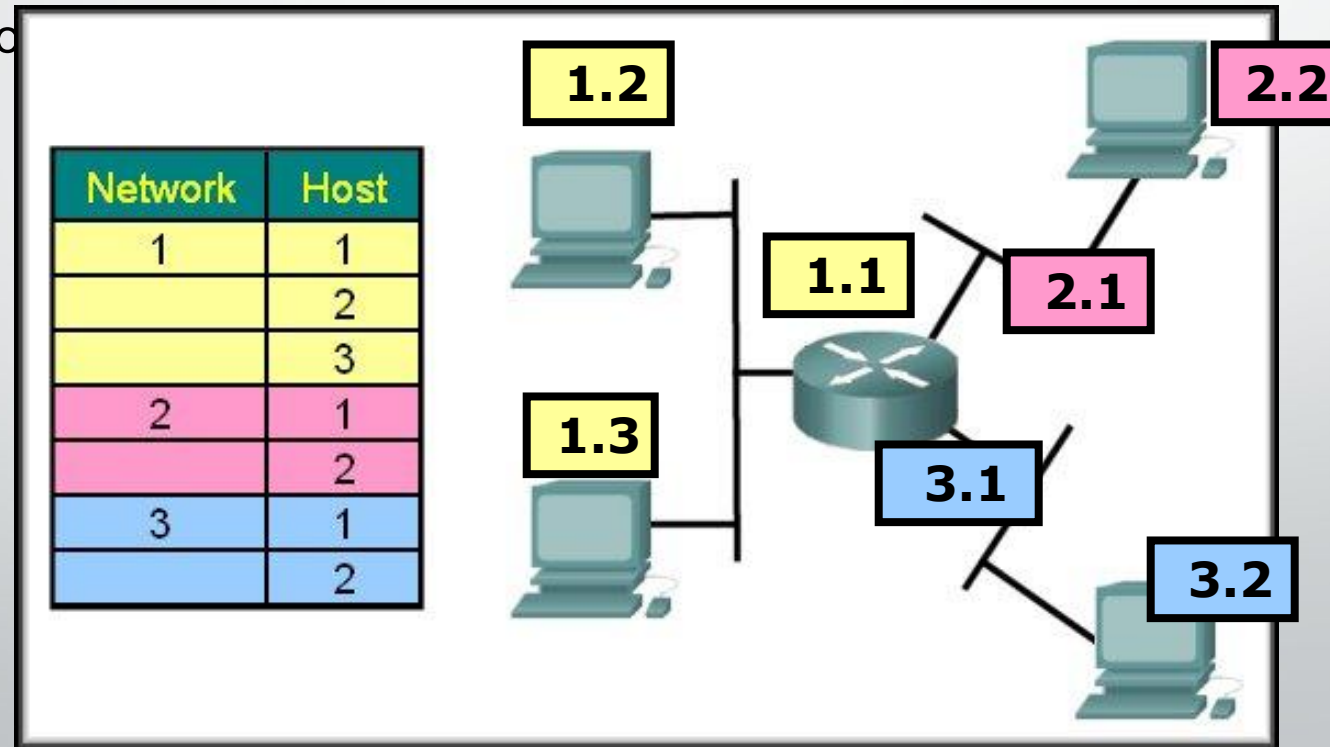
A 1 in this position means 64 is added to the total.

A 0 in any position means that 0 is added to the total.

11110101 in Binary = Decimal Number 245

Networks and Hosts

- To identify a path or "route" through a network, the address must be composed of two parts:
 - Network** portion
 - Host** portion



Network Portion

- Network Portion:**

- Some portion of the high-order bits
- A network can be defined as a group of hosts that have identical bit patterns in the network address portion of their addresses

IP Address	192.	168.	1.	2
Binary IP Address	11000000	10101000	00000001	00000010

192.168.1.2	11000000	10101000	00000001	00000010
192.168.1.67	11000000	10101000	00000001	01000011
192.168.1.204	11000000	10101000	00000001	11001100

Network Portion

- **Host Portion:**

- A variable number of least significant bits that are called the **host portion** of the address.
- The **number of bits** used in this **host portion** determines the **number of hosts** that we can

IP Address	192.	168.	1.	2
Binary IP Address	11000000	10101000	00000001	00000010

192.168.1.2	11000000	10101000	00000001	00000010
192.168.1.67	11000000	10101000	00000001	01000011
192.168.1.204	11000000	10101000	00000001	11001100

Prefix Mask

- How do we or devices identify the network part or the host part?

- Answer: Using the "Prefix Mask".

- 192.168.10.2/24

- Means that the first 24 bits are the network portion.
- The last 8 bits are the host portion.

- Subnet Mask is the other form of "Prefix Mask"

- Prefix length

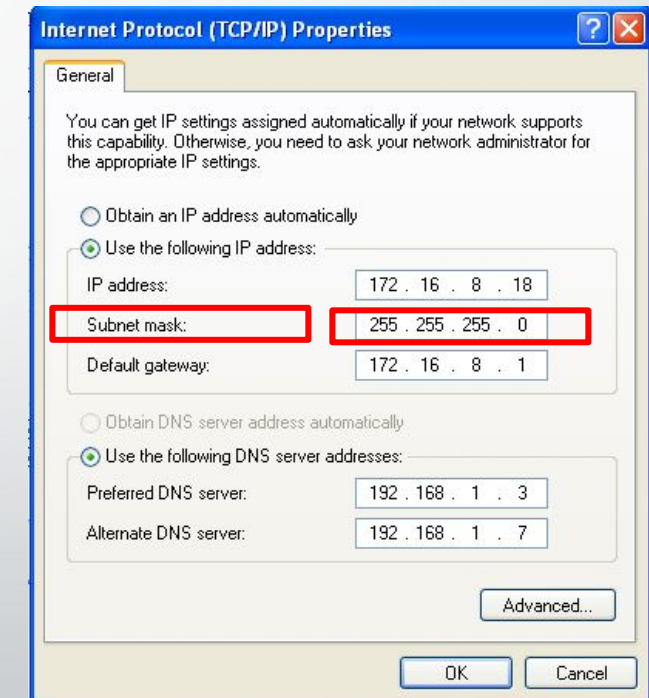
```
Z:\>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address                       . : 172.16.8.18
    Subnet Mask                       . : 255.255.255.0
    Default Gateway                   . : 172.16.8.1

Z:\>
```



Subnet Mask

- The Prefix Mask and the Subnet Mask are different ways of representing the **same information**.
- **Conversion:**
 - Subnet mask has the **same format** as an IP address. Hence, it has **32 bits divided into 8 bits (octets)**
 - Prefix mask of **/24** means, **the first (MSB) 24 bits** of subnet mask would be **1** and the rest will be **0**
 - Binary: **11111111.11111111.11111111.00000000**
 Decimal: **255.255.255.0**
 - Examples:
 - Prefix Mask of **/24** or a subnet mask of **255.255.255.0**
 - Prefix Mask of **/16** or a subnet mask of **255.255.0.0**
 - Prefix Mask of **/8** or a subnet mask of **255.0.0.0**

Exercise

- What's the **subnet mask** of the following?
 - IP Address: 10.24.36.2 / 4
 - IP Address: 10.24.36.2 / 12
 - IP Address: 10.24.36.2 / 16
 - IP Address: 10.24.36.2 / 23
- What's the **prefix mask** of the following?
 - IP Address: 10.24.36.2; Subnet Mask: 255.255.224.0
 - IP Address: 10.24.36.2; Subnet Mask: 255.255.255.192
 - IP Address: 10.24.36.2; Subnet Mask: 255.255.255.252
 - IP Address: 10.24.36.2; Subnet Mask: 255.254.0.0

ANDing the Binaries

- Inside data network devices, digital logic is applied for their interpretation of the addresses.
- AND is used in determining the network address.
 - $0 \text{ AND } 0 = 0$
 - $1 \text{ AND } 0 = 0$
 - $1 \text{ AND } 1 = 1$

	Decimal	Binary
IP Address	135.15.2.1	10000111 00001111 00000010 00000001
Subnet Mask	255.255.0.0	11111111 11111111 00000000 00000000
Network Address	135.15.0.0	

But Why AND?

- **Routers** use the **ANDing** process to determine the route a packet will take.
- The network number of the **destination IPv4 address** is used to find the network in the routing table.
- The router then determines the best path for the frame.

IPv4 Addresses

Network Address

Broadcast Address

Host Address

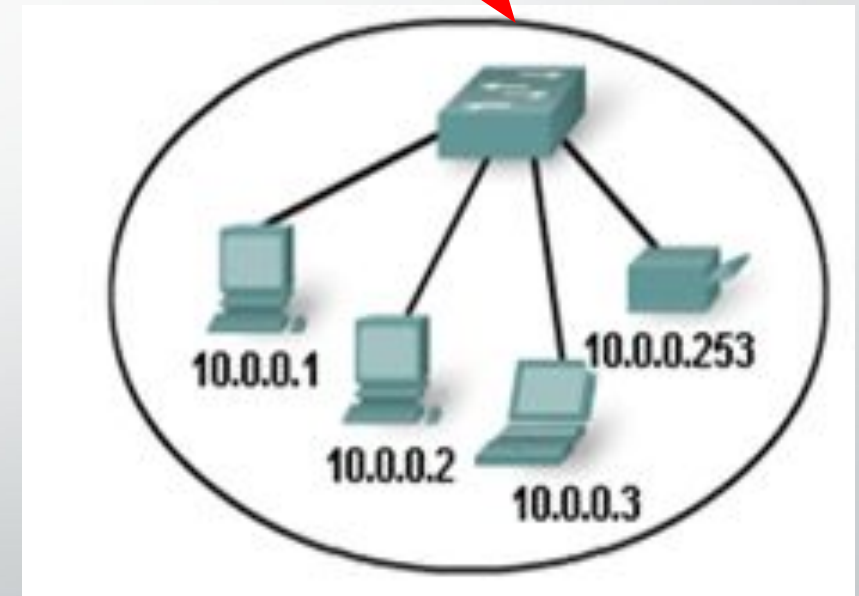
The Addresses

• Network Address

- All hosts in the network will have the same network bits.
- Cannot be assigned to a device.
- All host bits in this address will be **zero**.

IP Address	10.	0.	0.	0
IP Address in Binary	00001010	00000000	00000000	00000000
Network Part				Host Part

10.0.0.0/24



The Addresses

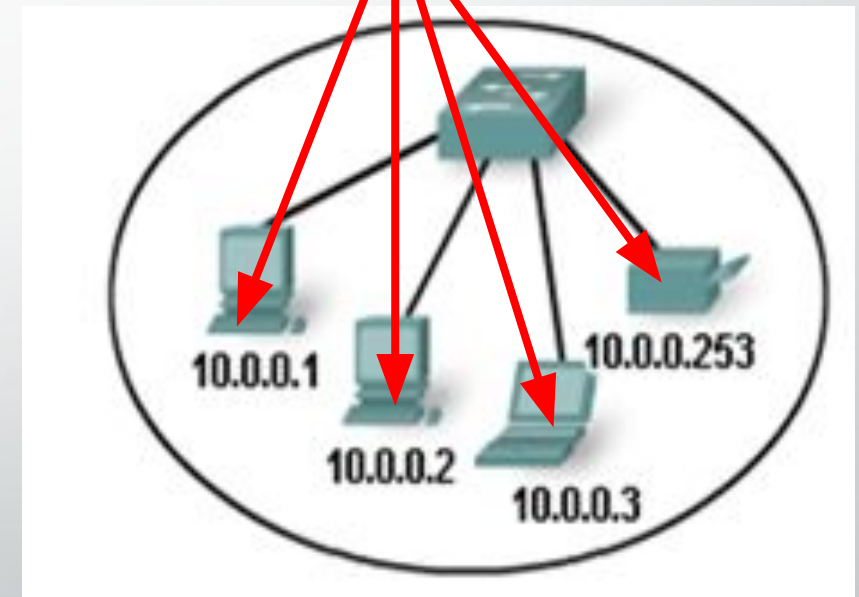
Broadcast Address

- Used to send message to all hosts in the network using one single address
- Cannot be assigned to a device.
- All host bits in this address will be **one**.

10.0.0.255/24

IP Address	10.	0.	0.	255
IP Address in Binary	00001010	00000000	00000000	11111111
Network Part				Host Part

Broadcast Address of 10.0.0.0/24 network is 10.0.0.255/24



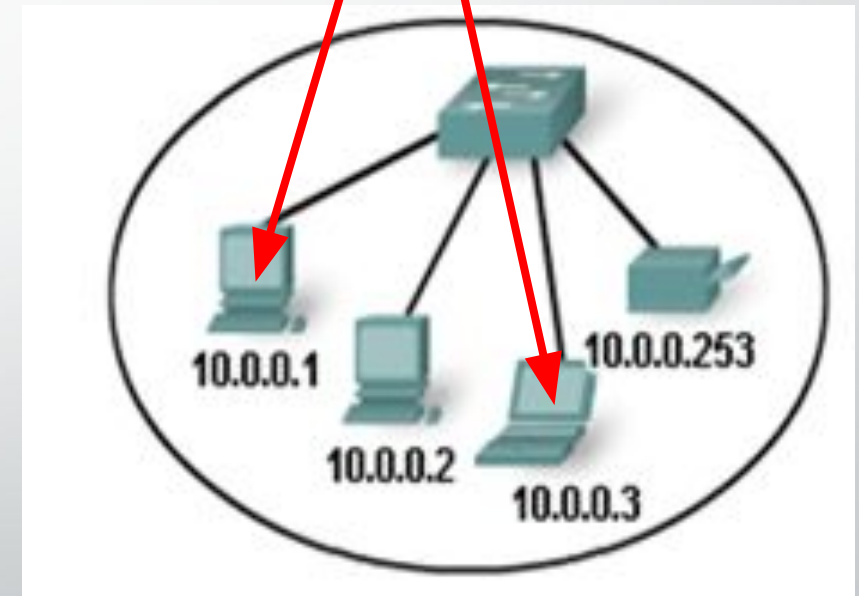
The Addresses

Host Address

- The unique address assigned to each device on the network.
- For a network of 10.0.0.0/24
 - Addresses 10.0.0.1 through 10.0.0.254 are
 - All host bits in this address will vary.

IP Address	10.	0.	0.	1
IP Address in Binary	00001010	00000000	00000000	00000001
Network Part				Host Part
IP Address	10.	0.	0.	3
IP Address in Binary	00001010	00000000	00000000	00000011

Host Address



The Addresses

- Say, you have a random IP address
192.168.10.193/24 or given as
192.168.10.193 255.255.255.0
- Say, you have a random IP address
200.32.16.192/26 or given as
200.32.16.192 255.255.255.192

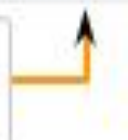
Network Prefix

- The network prefix is not always /24.

Using Different Prefixes for the 172.16.4.0 Network

Network	Network address	Host range	Broadcast address
172.16.4.0 /24	172.16.4.0	172.16.4.1 - 172.16.4.254	172.16.4.255
172.16.4.0 /25	172.16.4.0	172.16.4.1 - 172.16.4.126	172.16.4.127
172.16.4.0 /26	172.16.4.0	172.16.4.1 - 172.16.4.62	172.16.4.63
172.16.4.0 /27	172.16.4.0	172.16.4.1 - 172.16.4.30	172.16.4.31

**SAME NETWORK ADDRESS
ALL PREFIXES**

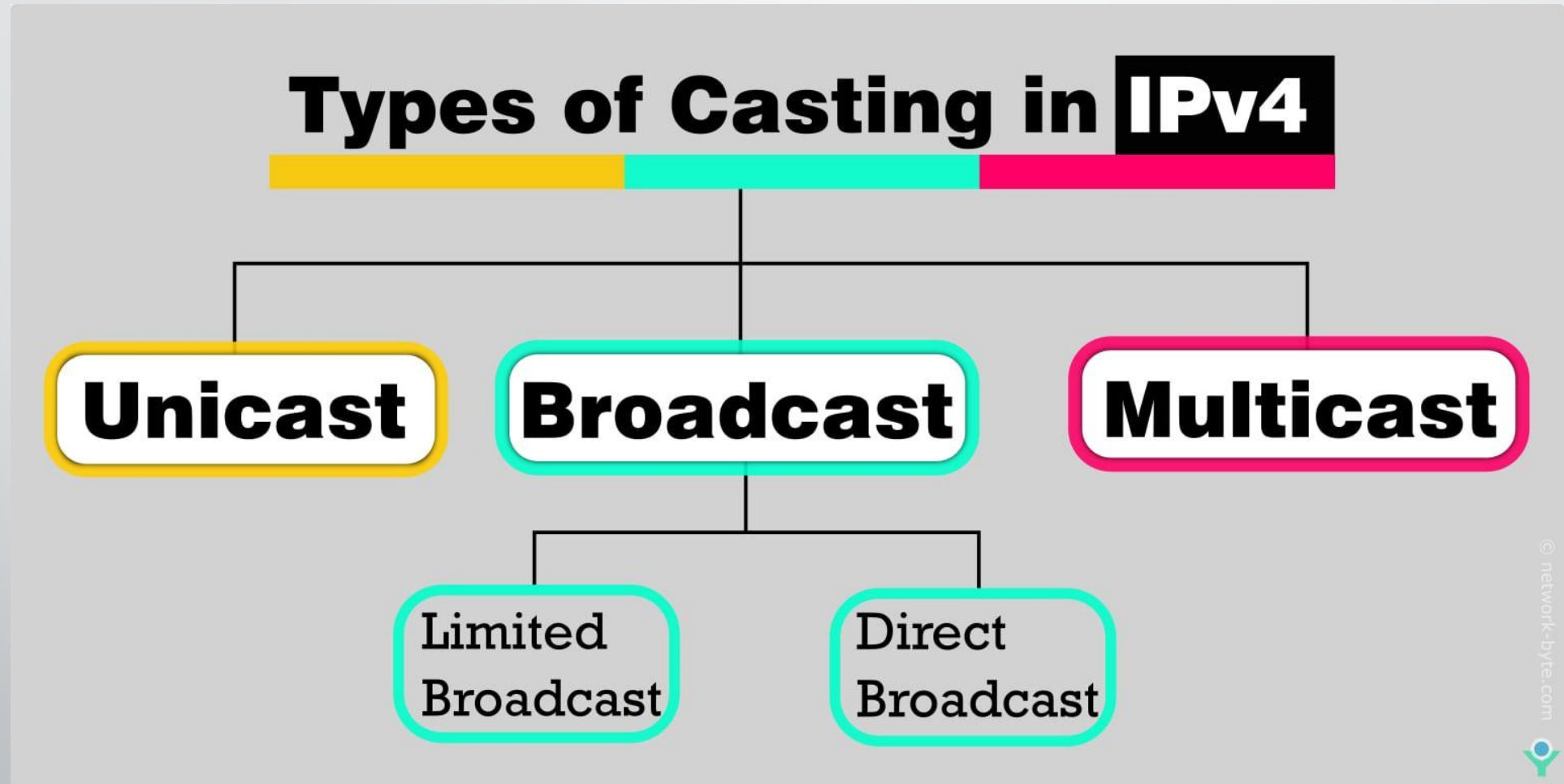


**DIFFERENT BROADCAST
ADDRESS EACH PREFIX**



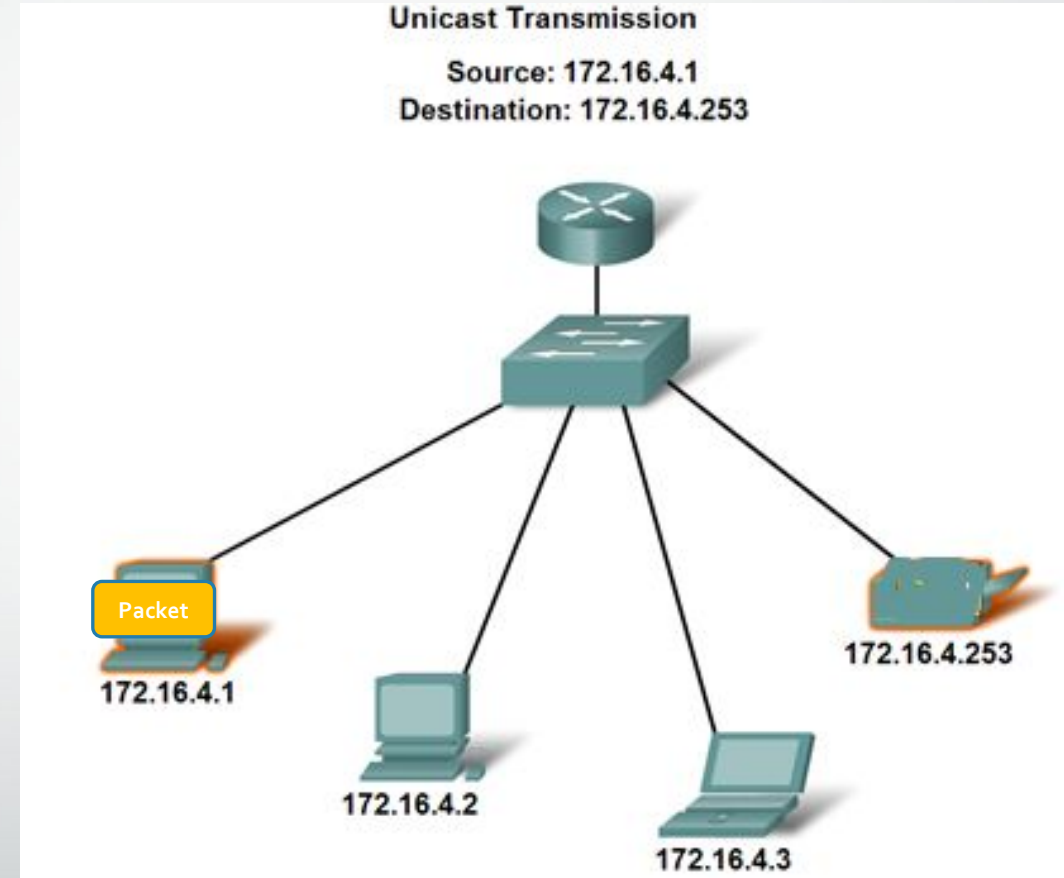
Types of IPv4 Addresses

Special Addresses



Unicast

- **Unicast**
 - A message to one host.
 - Individual IPv4 addresses



172.16.4.253

172.16.4.1

Data

Dest IP Add

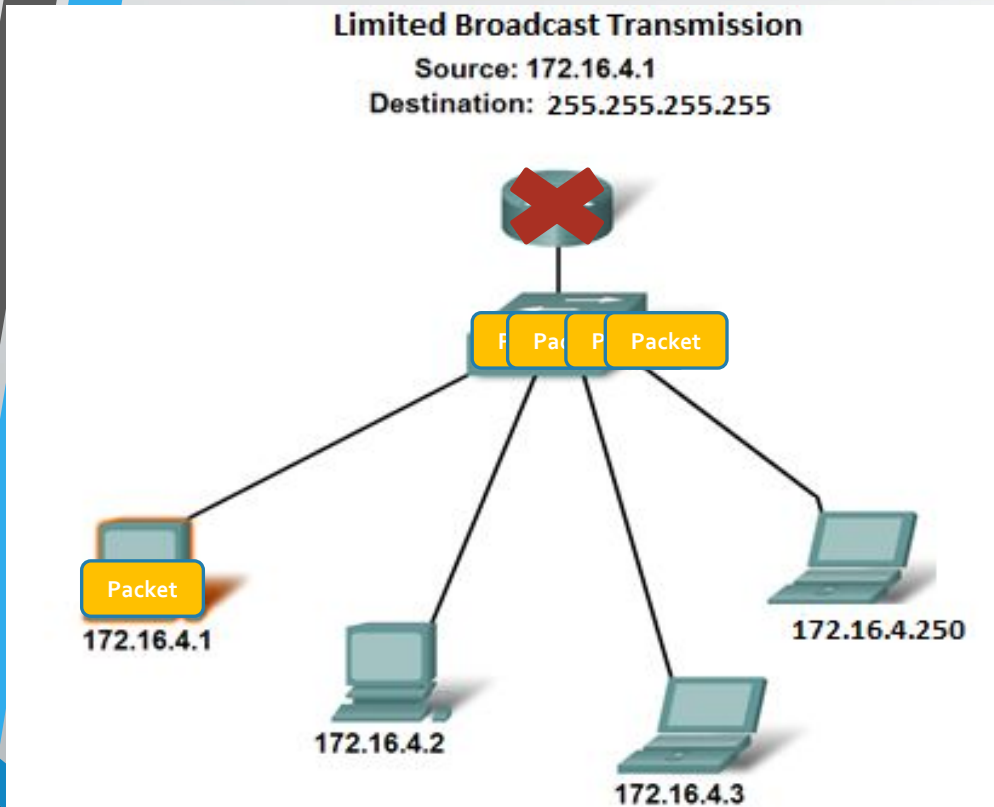
Source IP Add

Packet

Broadcast

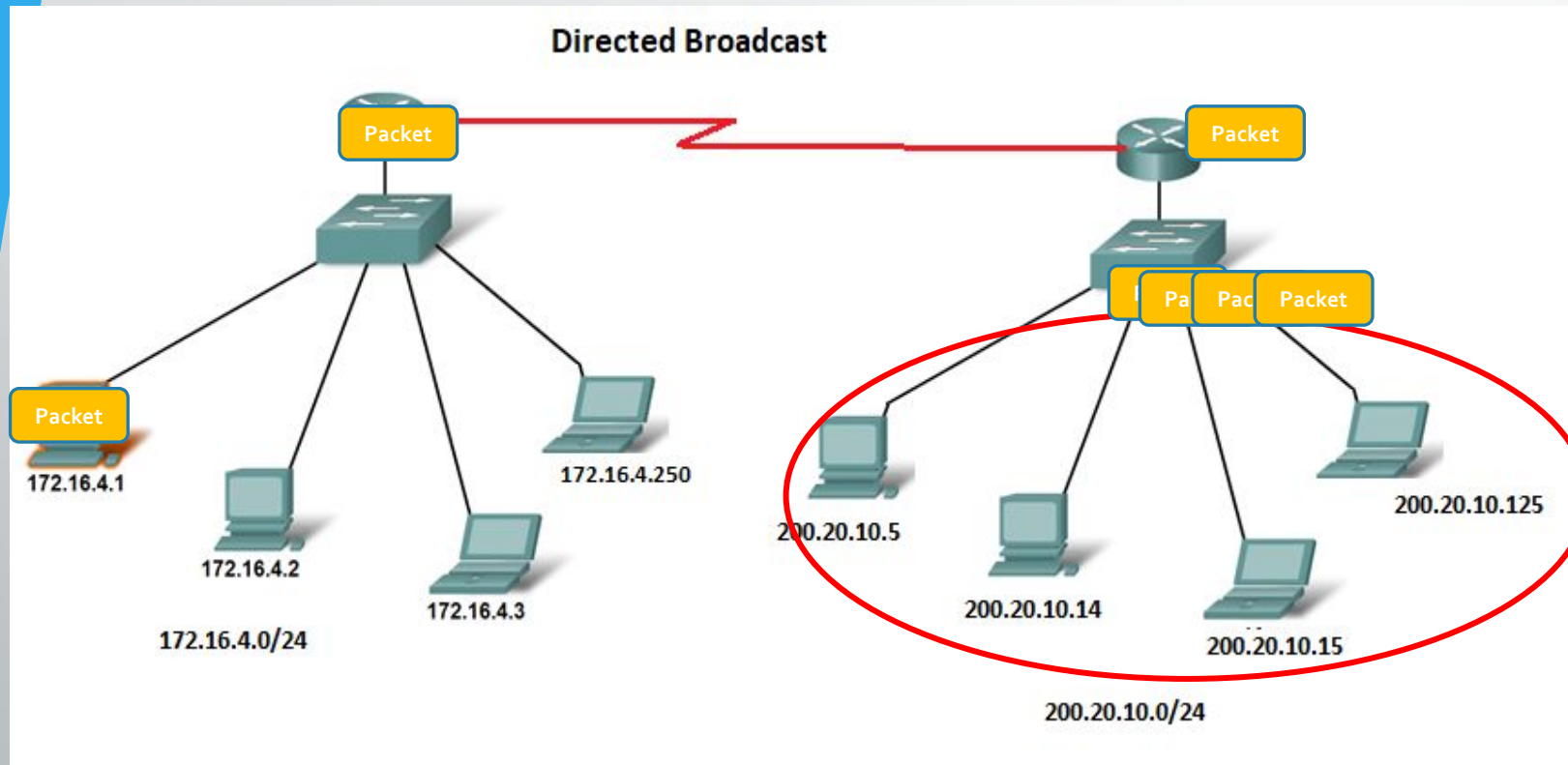
- **Limited Broadcast**

- A message to all hosts on the same physical/local network or subnet.
- **255.255.255.255**
- Never forwarded by routers!



255.255.255.255	172.16.4.1	Data
Dest IP Add	Source IP Add	Packet

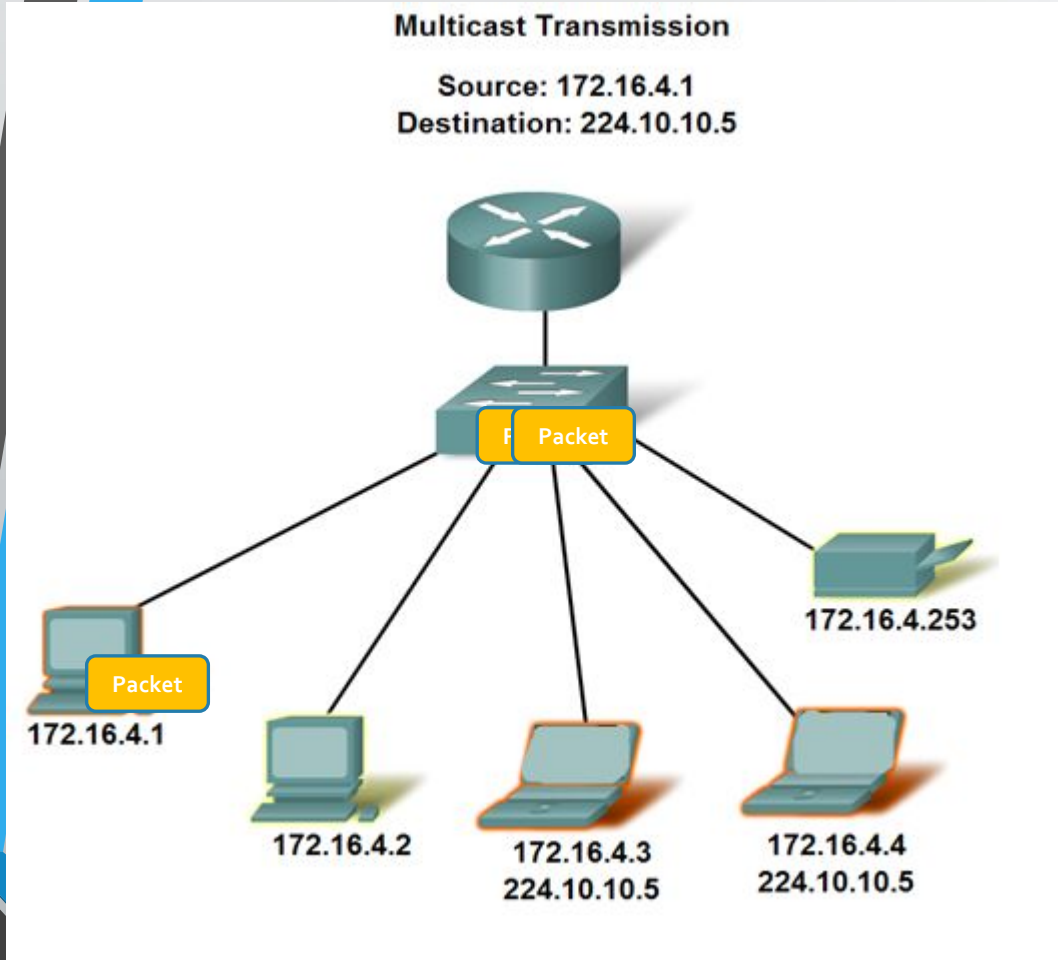
Broadcast



200.20.10.255	172.16.4.1	Data
Dest IP Add	Source IP Add	Packet

- **Directed Broadcast**
 - A message to all hosts on a different network or subnet.
 - **broadcast address of a network**
 - Example :
200.20.10.255

Multicast



224.10.10.5 172.16.4.1 Data

Dest IP Add Source IP Add Packet

• Multicast Addresses

- A message addressed to a group of hosts.
- Uses an IP address starting within this range of **224 - 239**

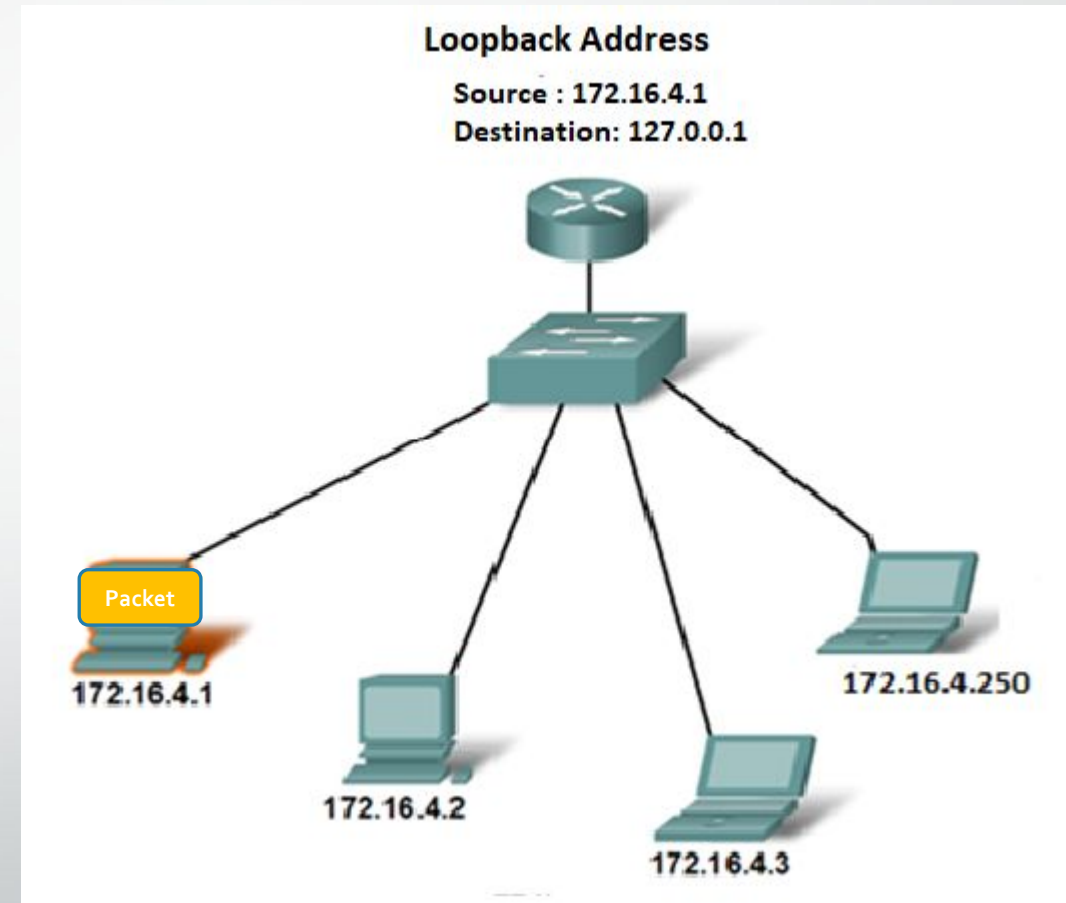
• Examples of Multicast Application

- Video and audio broadcasts
- Distribution of software
- News feeds

Loopback

• Loopback Address

- A message addressed to loop back in the device itself.
- **127.x.x.x** of **127.0.0.0/8**
- Not assigned to any device
- Testing and Troubleshooting purpose



127.0.0.1	172.16.4.1	Data
Dest IP Add	Source IP Add	Packet

Anatomy of IPv6 Address

Reasons for using IPv6

- **Address Availability:**

- **IPv4: 32 bits** - 4 octets

- 2^{32} or 4,294,467,295 IP Addresses.

- **IPv6: 128 bits** - 16 octets

- 2^{128}

- 3.4×10^{38} or

340,282,366,920,938,463,463,374,607,431,768,211,456

(340 undecillion) IP Addresses.



Every grain of sand on every beach on Earth could be assigned over a million unique IPv6 addresses, with plenty to spare (assuming approximately 7.5×10^{18} grains of sand globally and 2^{128} IPv6 addresses).

IPv6 Address

- 128 bits
- given below is a 128 bit IPv6 address represented in binary

```
0010000000000001 0000110110111000 1010110000010000 1111111000000001  
0000000000000000 0000000000000000 0000000000000000 0000000000000000
```

- Each 4 bits is converted into a Hexadecimal digit
- Each block contains 4 Hexadecimal digits

Each block is separated by ':' symbol

IPv6 Address

0010000000000001 0000110110111000 101011000010000 1111111000000001
0000000000000000 0000000000000000 0000000000000000 0000000000000000

2 0 0 1

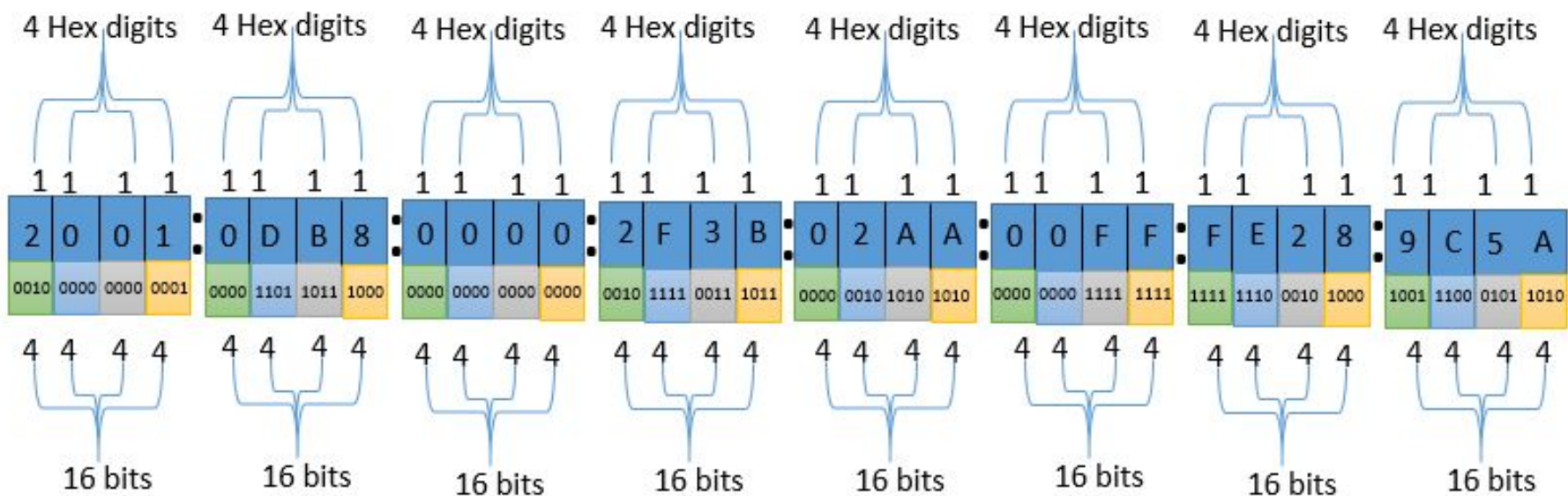
2001 : 0db8 : ac10 : fe01 : 0000 : 0000 : 0000 : 0000

- Called **string notation**

IPv6 Address

IPv6 address in Hexadecimal notation

2001:0DB8:0000:2F3B:02AA:00FF:FE28:9C5A



0010000000000000100001101101110000000000000000000001011110011101100000010101010100000000111111111111110001010001001110001011010

IPv6 address in binary notation

IPv6 Addressing

- IPv6 Representation – Rule 1:

- The leading zeros in any 16-bit segment do not have to be written. If any 16-bit segment has fewer than four hexadecimal digits, it is assumed that the missing digits are leading zeros.

2031 : 0000 : 130F : 0000 : 0000 : 09C0 : 876A : 130B

2031 : 0 : 130F : 0 : 0 : 9C0 : 876A : 130B

8105 : 0000 : 0000 : 4B10 : 1000 : 0000 : 0000 : 0005

8105 : 0 : 0 : 4B10 : 1000 : 0 : 0 : 5

0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000

0 : 0 : 0 : 0 : 0 : 0 : 0 : 0

IPv6 Addressing

- **IPv6 Representation – Rule 2:**
 - Any single, contiguous string of one or more 16-bit segments consisting of all zeroes can be represented once with a double colon.

1080:0:0:0:8:800:200C:417A =
FF01:0:0:0:0:0:0:101 =
0:0:0:0:0:0:0:1 =
0:0:0:0:0:0:0:0 =

IPv6 Addressing

- IPv6 Representation – Rule 2:

- *Any single, contiguous string of one or more 16-bit segments consisting of all zeroes can be represented once with a double colon.*

Example: **1843::22::fa**

- Illegal because the length of the two all-zero strings is ambiguous.

1843:0000:0000:0000:0022:0000:0000:00fa

or

1843:0000:0000:0022:0000:0000:0000:00fa

Representing IPv6 addresses



- IPv4 Address



- IPv6 Address
- No Subnet masks in dotted decimal format in IPv6

The End